Diabetes Mellitus CBME Training Module Objectives

Learning Objectives - Competency-Based

By the end of this 2-week module, MBBS students will be able to:

🎯 Patient Care Competencies (PC)

PC1: Community Screening & Diagnosis

- **Competency Statement**: Conduct comprehensive diabetes screening in community settings using evidence-based protocols
- **Observable Behaviors**:
  - Perform capillary blood glucose testing using glucometer
  - Interpret random blood glucose, fasting plasma glucose, and HbA1c results
  - Identify patients requiring further diagnostic testing (OGTT)
  - Document screening results accurately in patient records

PC2: Patient Education & Counseling

- **Competency Statement**: Provide culturally appropriate diabetes education and lifestyle counseling to patients and families
- **Observable Behaviors**:
  - Explain diabetes pathophysiology in simple, understandable language
  - Demonstrate dietary modifications for glycemic control
  - Teach self-monitoring blood glucose techniques
  - Counsel on physical activity recommendations for diabetes management

PC3: Multidisciplinary Care Coordination

- **Competency Statement**: Coordinate care between primary care, specialists, and community resources for diabetic patients
- **Observable Behaviors**:
  - Identify referral indications for endocrinology consultation
  - Collaborate with ASHAs for follow-up care
  - Coordinate pharmaceutical management with community pharmacists
  - Arrange home visits for non-compliant patients

PC4: Complication Assessment

- **Competency Statement**: Perform basic screening for diabetes complications and provide preventive care
- **Observable Behaviors**:
  - Conduct foot examination for neuropathy and vascular changes
  - Assess visual acuity and arrange eye screening
  - Evaluate cardiovascular risk factors

- Screen for diabetic nephropathy using urine dipstick

🧠 Medical Knowledge Competencies (MK)

MK1: Pathophysiology Understanding

- **Competency Statement**: Explain the pathophysiology of type 1, type 2, and gestational diabetes mellitus
- **Knowledge Elements**:
  - Beta cell dysfunction in type 2 diabetes
  - Insulin resistance mechanisms
  - Glycemic index and carbohydrate metabolism
  - Acute and chronic complications pathways

MK2: Risk Factors & Prevention

- **Competency Statement**: Identify modifiable and non-modifiable risk factors and implement prevention strategies
- **Knowledge Elements**:
  - Lifestyle modification for diabetes prevention
  - Genetic predisposition and familial risk
  - Gestational diabetes screening protocols
  - Metformin for diabetes prevention in high-risk groups

MK3: Management Guidelines

- **Competency Statement**: Apply evidence-based guidelines for diabetes management in Indian context
- **Knowledge Elements**:
  - ADA/IDF treatment algorithms
  - NCDIR guidelines for Indian population
  - Medication selection based on patient profile
  - Glycemic, BP, and lipid targets

MK4: Emergency Recognition

- **Competency Statement**: Recognize and manage diabetes-related emergencies in community settings
- **Knowledge Elements**:
  - Hypoglycemia management without IV access
  - Hyperglycemic hyperosmolar state recognition
  - Diabetic ketoacidosis initial management
  - Referral protocols for emergencies

📚 Practice-Based Learning Competencies (PBLI)

PBLI1: Evidence-Based Approach

- **Competency Statement**: Apply evidence-based medicine principles to diabetes care decisions
- **Observable Behaviors**:

    - Search and critically appraise diabetes research
    - Apply clinical practice guidelines to patient care
    - Participate in quality improvement initiatives
    - Present case studies with evidence-based recommendations

PBLI2: Self-Directed Learning

- **Competency Statement**: Engage in self-directed learning through patient encounters and reflective practice
- **Observable Behaviors**:
    - Maintain learning portfolio with diabetes cases
    - Seek feedback on clinical performance
    - Set personal learning goals based on patient care experiences

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💬 Communication Skills Competencies (CS)

CS1: Patient-Provider Communication

- **Competency Statement**: Communicate effectively with diabetic patients from diverse backgrounds
- **Observable Behaviors**:
    - Use clear, non-technical language for patient education
    - Demonstrate empathy in difficult conversations
    - Employ teach-back method for education verification
    - Respect cultural beliefs while providing evidence-based care

CS2: Family & Community Engagement

- **Competency Statement**: Engage families and community leaders in diabetes prevention and management
- **Observable Behaviors**:
    - Conduct family counseling sessions
    - Collaborate with school teachers for childhood obesity prevention
    - Work with local leaders for community-based interventions
    - Organize support group meetings for diabetic patients

CS3: Interdisciplinary Communication

- **Competency Statement**: Communicate effectively with healthcare team members and other professionals
- **Observable Behaviors**:
    - Present patient cases in multidisciplinary meetings
    - Document handovers clearly using SBAR format
    - Collaborate with nutritionists and physiotherapists
    - Coordinate with pharmacists for medication counseling

🧑‍⚕️ Professionalism Competencies (PRO)

PRO1: Ethical Practice

- **Competency Statement**: Practice medicine ethically with particular attention to diabetes care
- **Observable Behaviors**:
  - Obtain informed consent for screening procedures
  - Maintain patient confidentiality during community activities
  - Respect patient autonomy in treatment decisions
  - Advocate for equitable access to diabetes care

PRO2: Cultural Competence

- **Competency Statement**: Provide culturally sensitive diabetes care to diverse populations
- **Observable Behaviors**:
  - Adapt education materials for local languages and literacy levels
  - Respect traditional healing practices while promoting evidence-based care
  - Recognize socioeconomic determinants of diabetes management

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🏥 Systems-Based Practice Competencies (SBP)

SBP1: Healthcare System Navigation

- **Competency Statement**: Navigate the Indian healthcare system to optimize diabetes care delivery
- **Observable Behaviors**:
  - Guide patients through NPCDCS (National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke)
  - Access government schemes for diabetes medication and supplies
  - Coordinate between primary, secondary, and tertiary care levels

SBP2: Resource Management

- **Competency Statement**: Optimize resource utilization in diabetes care management
- **Observable Behaviors**:
  - Select cost-effective diagnostic and treatment options
  - Prioritize high-risk patients for limited resources
  - Participate in community resource mobilization

SBP3: Quality Improvement

- **Competency Statement**: Participate in quality improvement initiatives for diabetes care
- **Observable Behaviors**:
  - Analyze community diabetes screening data
  - Develop improvement plans for diabetes care processes
  - Monitor and evaluate intervention effectiveness

📊 Assessment Methods

Formative Assessment:

- **Portfolio Review**: Weekly documentation of patient encounters
- **Skills Checklist**: Observed performance of clinical competencies
- **Peer Assessment**: Group work and presentation evaluations
- **Self-Reflection**: Daily learning logs and competency self-assessment

Summative Assessment:

- **OSCE Stations**: Testing clinical skills in simulated scenarios
- **Portfolio Defense**: Presentation of comprehensive patient care experience
- **Written Examination**: Testing theoretical knowledge and clinical decision-making
- **Community Project**: Evaluation of group intervention planning and execution

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🎯 Exit Criteria - Mastery Level

Students must demonstrate competency at level 3 or 4 (shows/supervision required to does/autonomous performance) on the following scale:

1. **Knowledge** (Aware of facts/principles)
2. **Comprehension** (Understands facts/relationships)
3. **Application** (Applies knowledge in familiar situations)
4. **Analysis** (Breaks information into components)
5. **Synthesis** (Integrates information into new understanding)
6. **Evaluation** (Makes judgments based on criteria)

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